Counters and Registers

Counters: An electronic digital counter is a flip flop circuit capable of counting the number of clock pulses applied to it. It produces binary output in sequence. Counter is a sequential circuit. It is a group of flip-flops with a clock signal applied. The total number of states produced by counter is denoted by 'M' it specifies modulus of counter. The required no. of flip flops can be calculated by formula

Mod of counter = M = 2^n where n= no. of flip flops.

for example : For mod 16 counter 2^n=16 i.e. 4 flip flops are required.

Counters are classified as Asynchronous and Synchronous counters.

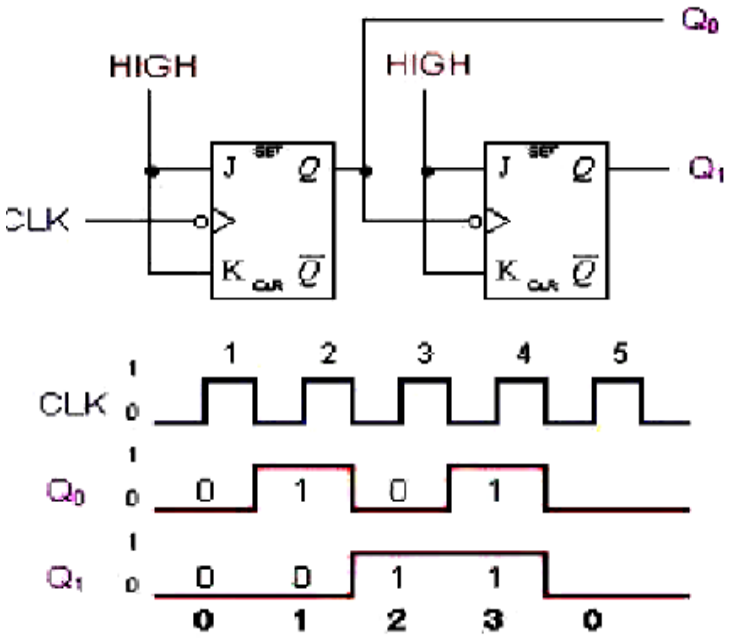
Asynchronous Counters:

Asynchronous counters are those counters which **do not operate on simultaneous clocking**. In asynchronous counter, only the first flip-flop is externally clocked using clock pulse while the clock input for the successive flip-flops will be the output from a previous flip-flop.

This means that only a single clock pulse is not driving all the flip-flops in the arrangement of the counter. Asynchronous counters are also known as ripple counters and are formed by the successive combination of trailing edge-triggered flip-flops. It is called so because the data ripples between the output of one flip-flop to the input of the next. Counters may be up counter, down counter and up-down counter. For the design of the asynchronous counter, T flip-flops are used. Because the output toggles in T flip-flop. In other words, this flip-flop produces complementing output. That is, if 0 is given as the input, 1 is produced at the output and vice versa.

2 bit Binary UP counter :

For a 2-bit asynchronous counter, 2 flip-flops are used as shown in the below figure. It is an up counter, where the count is incremented for each clock pulse. A 2-bit counter will have $2^2 = 4$ distinct states(00, 01, 10, 11).



Clock Pulse	Q_1	Q_0
1	0	0
2	0	1
3	1	0
4	1	1

Figure 4.56 Two bit binary Ripple Up Counter with wave form & count table

The clock pulse input is given to the first flip-flop only. The output(Q_0) of the first flip-flop is fed as the clock input for the second flip-flop. The HIGH or logic 1 is maintained at the T-input of both flip-flops.

The clock pulse is digitally counted at the output Q_0 and Q_1, where Q_0 is the Least significant bit(LSB) and Q_1 is the most significant bit(MSB).

Operation

Let us assume the initial state of flip-flop as $Q_1Q_0 = 00$. During the first clock pulse, that is at the falling edge of the clock pulse, JK FF1 will toggle its output state to logic 1. For the next clock pulse, again JK FF1 will toggle its output to logic 0. The toggling of JK FF1 continues for each clock pulse.

Now, *What about the state of JK FF2 ?*

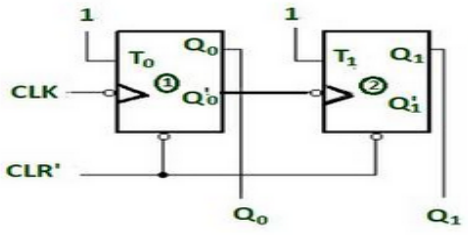
The output of the first flip-flop(JK FF1) is given as a clock pulse input for the second flip-flop(JK FF2). Whenever the Q_0 output goes high, at the falling edge of its output, the JK FF2 will get triggered. When the JK FF2 gets triggered, it produces a toggled output.

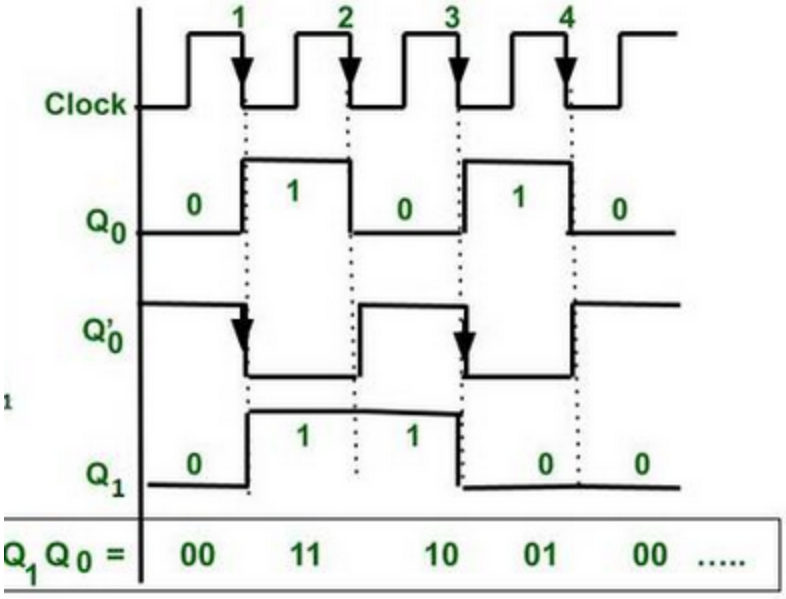
As we have assumed the initial state of JK FF2 as $Q_1 = 0$, the output will get toggled to $Q_1 = 1$. This output again toggles, when the output of JK FF1 goes high HIGH.

The change of state of flip-flops for each occurrence of the clock pulse is shown in the timing diagram. As you can observe from the timing diagram, the counter counts from 00 to 11. After reaching 11, the counter resets to 00 and starts to count from 00 again.

2 bit binary down counter

For a 2-bit asynchronous counter, 2 flip-flops are used as shown in the below figure. It is an down counter, where the count is decremented for each clock pulse. A 2-bit counter will have $2^2 = 4$ distinct states(00,11, 10, 01,00).





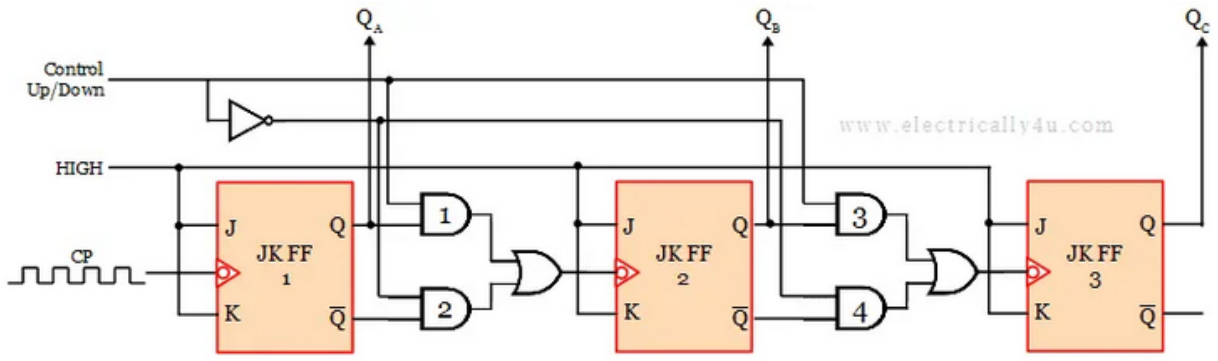
The clock pulse input is given to the first flip-flop only. The output(Q_0') of the first flip-flop is fed as the clock input for the second flip-flop. The HIGH or logic 1 is maintained at the T-input of both flip-flops.

The clock pulse is digitally counted at the output Q_0 and Q_1, where Q_0 is the Least significant bit(LSB) and Q_1 is the most significant bit(MSB).

Asynchronous Up/Down counter (or) Bidirectional counter

For asynchronous up/down counter, control input is necessary for the control of the operation as up or down counter.

When the control input is 0, in the following figure, the AND gates 1 and 3 are disabled. It enables gate 2 and 4, which means, Q' output of first and second flip-flop drives the clock input of JK FF2 and JK FF3 respectively. In this case, the counter will perform the operation of the down counter.



When the control input is 1, gates 1 and 3 are enabled and gates 2 and 4 are disabled. This makes a way for the Q output of flip-flops to drive the clock input of the next stage flip-flop. Now, the counter will perform the up counter operation.

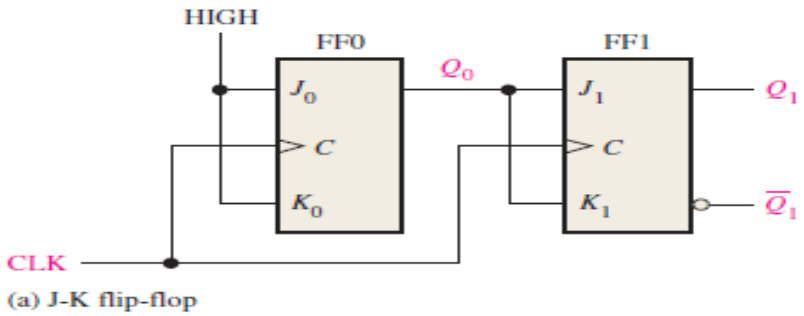
Synchronous Counter:

When the counter is clocked such that each flip flop in the counter is triggered at the same time, the counter is called as synchronous counter. Design of these counters involve complex logic circuit as no. of states increases. As clock is given simultaneously to all flip flops there is no problem of propagation delay . Hence they are high speed counters and are preferred when no. of flip flops increases in the given design.

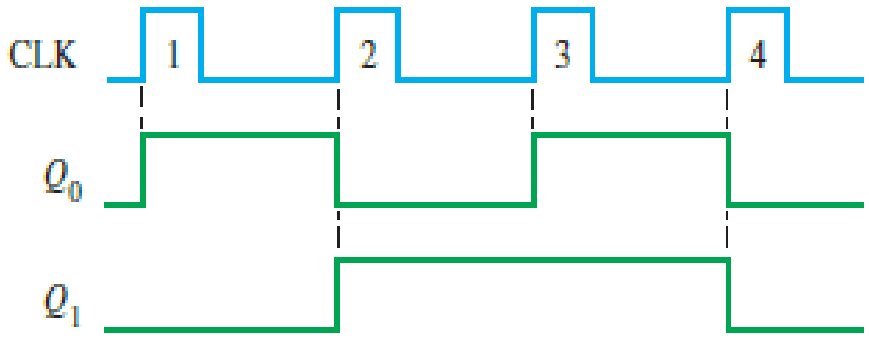
2 bit UP counter:

The J_0 and K_0 inputs of FF-0 are tied to logic 1. So FF-0 will work as a toggle flip-flop. The J_1 and K_1 inputs are connected to Q_0.

Logic diagram:



(a) J-K flip-flop



Clock Pulse	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4 (recycles)	0	0

Working :
Initially Q_1Q_0=00
As soon as the first positive clock edge is applied, FF-0 will toggle and Q_0 will change from 0 to 1.But at the instant of application of positive clock edge, Q_0 , $J_1 = K_1 = 0$. Hence FF-1 will not change its state. So Q_1 will remain 0.

Q_1Q_0 = 01 after the first clock pulse.

On the arrival of second positive clock edge, FF-0 toggles again and Q_0 changes from 1 to 0.But at this instant Q_0 was 1. So $J_1 = K_1 = 1$ and FF-1 will toggle. Hence Q_1 changes from 0 to 1.

Q_1Q_0 = 10 after the second clock pulse.

On application of the third positive clock edge, FF-0 will toggle from 0 to 1 but there is no change of state for FF-1.

Q_1Q_0 = 11 after the third clock pulse.

On application of the next clock pulse, Q_0 will change from 1 to 0 as Q_1 will also change from 1 to 0.

Q_1Q_0 = 00 after the fourth clock pulse.

Synchronous up/down cou

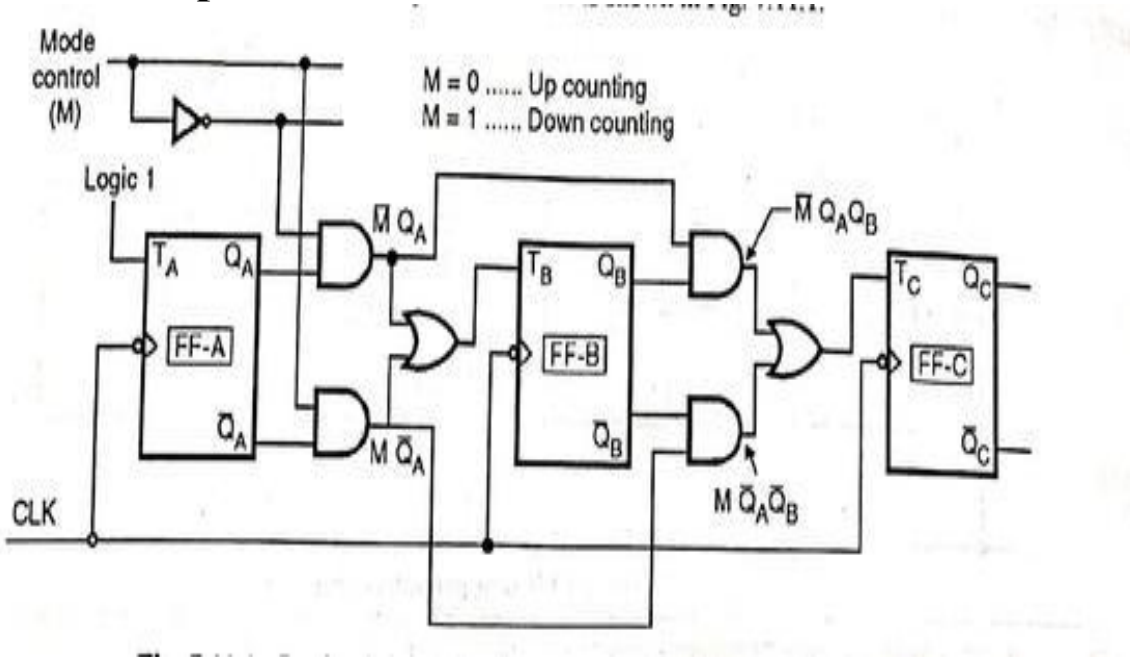


Fig shows 3 bit up down counter that will count from 000 to 111 when the M=0 and will count from 111 to 000 when M=1.

When the control input is 0, in the figure, the lower AND gates are disabled. It enables upper AND gates which means, Q output of first and second flip-flop drives the clock input of JK FFB and JK FFC respectively. In this case, the counter will perform the operation of the up counter. When the control input is 1, Upper gates are disabled and lower gates are enabled. This makes a way for the Q' output of flip-flops to drive the clock input of the next stage flip-flop. Now, the counter will perform the down counter operation.

Applications

- For suppose, in our kitchen appliances, we use microwave ovens. In that we set some temperature to heat the food item kept in it. Internally the counter calculates the increase or decrease in temperature and time. If it reaches the pre-set temperature, then it prevents from further heating and spoiling of that food item.

- Washing machines: We use counters in washing machines also. Similar to the counting operation in microwave oven, the counter in washing machine counts the time which we set it to operate.

- In both microwave oven and washing machine, we set the device to particular time, and it starts decreasing for every second. When the value of counter becomes zero, it activates the switch ON / OFF. Thus the operation of the device is controlled by counters.

Registers

Register is a group of flip flops used to store multiple bits of data. A flip flop can store 1 bit information. So an n bit register has group of n flip flops and is capable of storing any binary information /number containing n bits.

Shift Registers

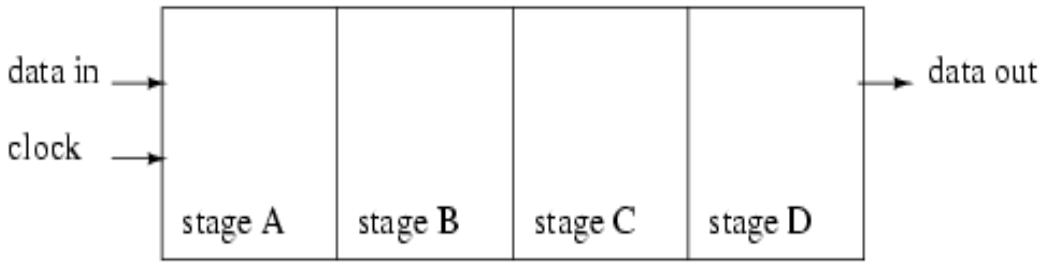
The binary information (data) in a register can be moved from stage to stage within the register or into or out of the register upon application of clock pulses. This type of bit movement or shifting is essential for certain arithmatic and logic operations used in microprocessors.This gives rise to group of registers called shift registers.They are very important in applications involving the storage and transfer of data in digital system.

Modes of operation of shift registers

- Serial In Serial Out shift register
- Serial In parallel Out shift register
- Parallel In Serial Out shift register
- Parallel In parallel Out shift register

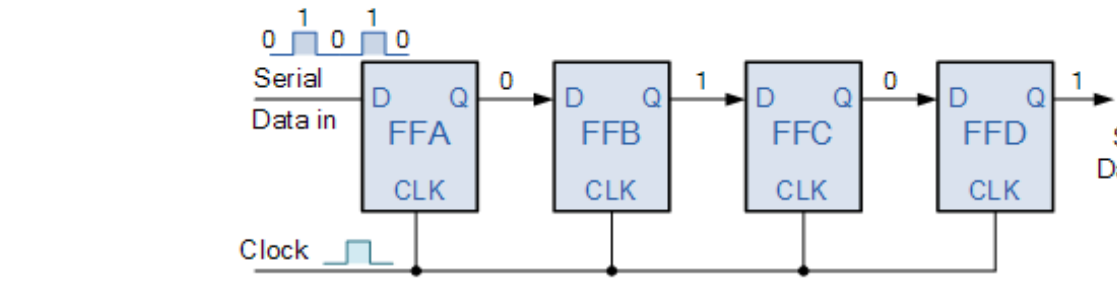
Serial In Serial out shift register

- The input to this register is given in serial fashion i.e. one bit after the other through a single data line and the output is also collected serially.
- The data can be shifted only left or shifted only right. Hence it is called Serial in Serial out shift register or a SISO shift register.



Serial-in, serial-out shift register with 4-stages

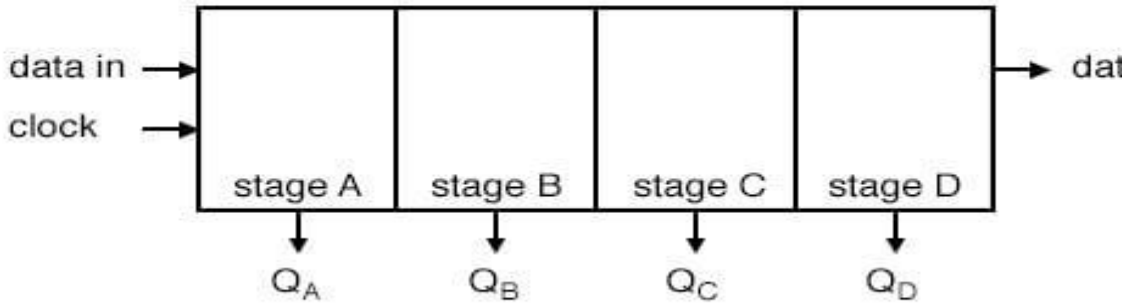
A *shift register* basically consists of several single bit "D-Type Data Latches", one for each data bit, either a logic "0" or a "1", connected together in a serial type daisy-chain arrangement so that the output from one data latch becomes the input of the next latch and so on.



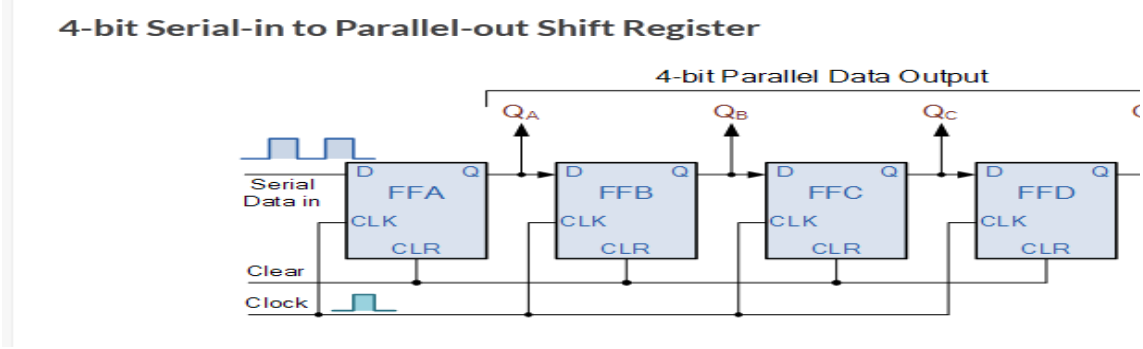
Total no. of clock cycles required are 2n-1 where n = no. of bits

Serial In Parallel Out

- The input to this register is given in serial and the output is collected in parallel**.**



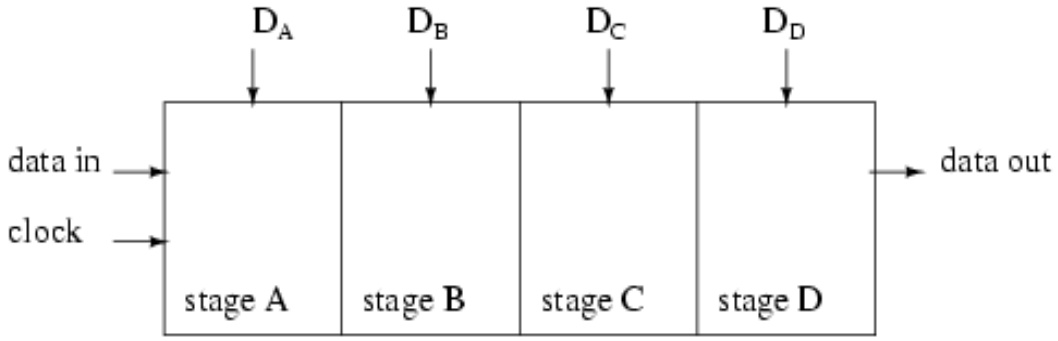
Serial-in parallel-out shift register with 4-stages



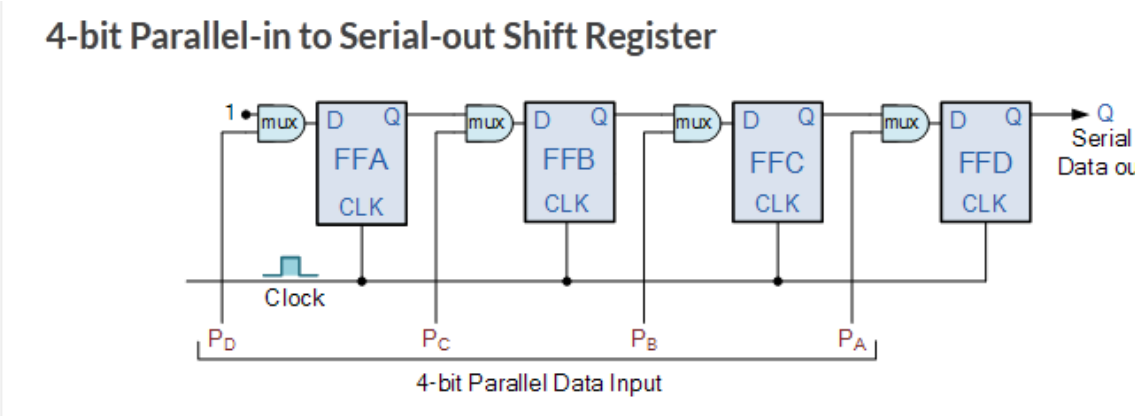
Number of clock pulses required are n

Parallel In Serial Out

The input to this register is given in parallel i.e. data is given separately to each flip flop and the output is collected in serial at the output of the end flip flop**.**



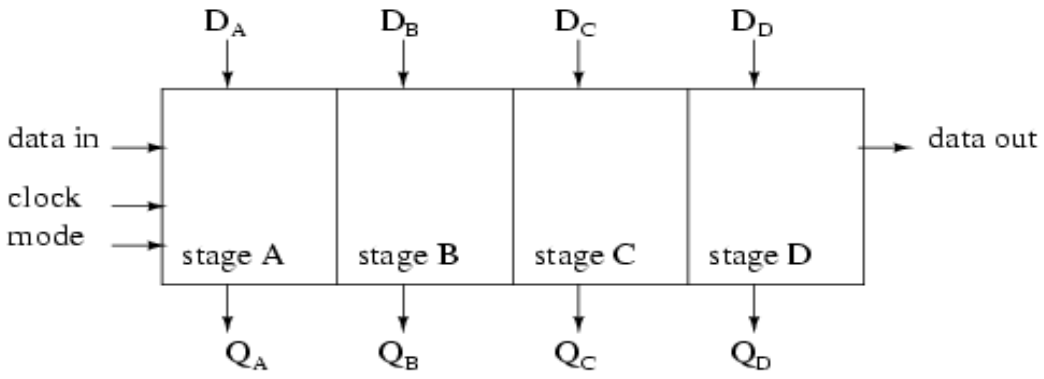
Parallel-in, serial-out shift register with 4-stages



No. of clock cycles are n

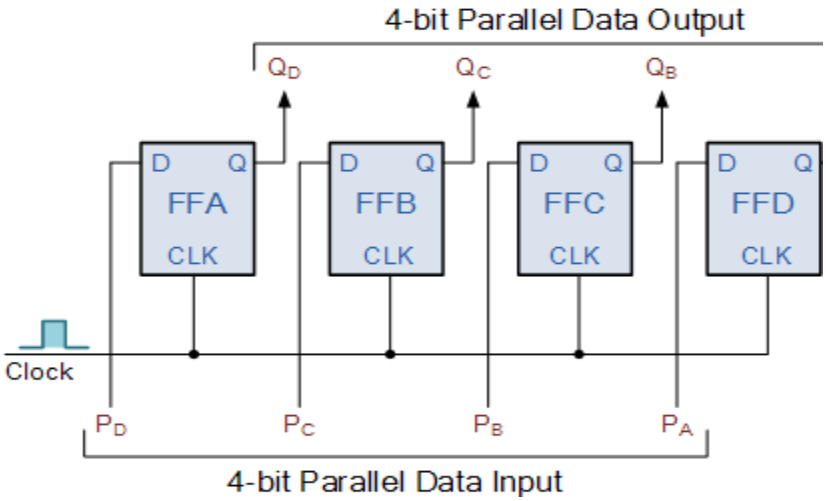
Parallel In Parallel OUT

- In this register, the input is given in parallel and the output also collected in parallel.
- Clock signals are connected to all the 4 flip flops. Data is given as input separately for each flip flop and in the same way, output also collected individually from each flip flop.



Parallel-in, parallel-out shift register with 4-stages

4-bit Parallel-in to Parallel-out Shift Register



No of clock cycles only1

Applications

1.Delay Line

2. Serial to Parallel Converter

3. Parallel to Serial Converter

4 Temporary data storage

5.Data Transfer